

HI × UGRB Cross-Correlation Pipeline

Complete Physics Reference

Research Progress Update

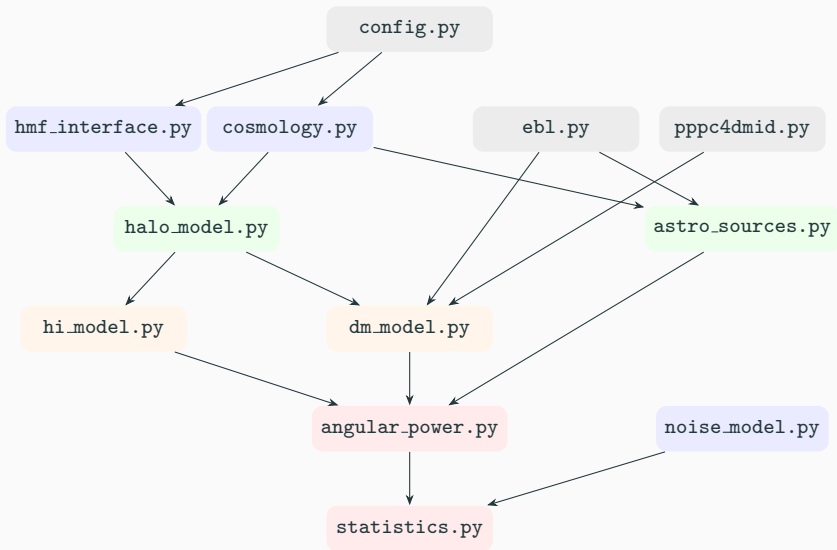
April 23, 2026

Research Overview

Hypothesis: DM traces LSS and annihilation events from particle-like DM has gamma-ray SED detectable from $C_{\ell}^{\text{HI} \times \text{DM}}$, subdominant in $C_{\ell}^{\text{HI} \times \gamma}$.

- Cross-correlate **HI 21-cm intensity maps** (MeerKAT) with the **unresolved gamma-ray background** (Fermi-LAT)
- Detect astrophysical cross-signal $C_{\ell}^{\text{HI} \times \star}$ (blazars, mAGN, SFGs sharing LSS with HI)
- Constrain dark matter annihilation via excess $C_{\ell}^{\text{HI} \times \gamma}$
- Based on Pinetti, Camera, Fornengo & Regis (2020), arXiv:1911.04989

Module Dependency DAG



Unit Conventions

Quantity	Unit	Convention
Distances	Mpc/ h (comoving)	χ , R_{vir} , r_s
Masses	$M_{\odot}/h$	M , M_{HI}
Wavenumbers	h/Mpc	k
Power spectra	$(\text{Mpc}/h)^3$	P_{lin} , P_{HI}
Mass function	$(\text{Mpc}/h)^{-3} (M_{\odot}/h)^{-1}$	dn/dM
Brightness temp.	mK	$\bar{T}_b$
Luminosities	erg/s	GLFs
Fermi noise	$\text{cm}^{-4} \text{s}^{-2} \text{sr}^{-1}$	N^{γ}

All internal calculations use h -dependent comoving units.

Physical units appear only at module boundaries (explicit conversion).

Cosmology

Planck 2018 Parameters

$$H_0 = 67.36 \text{ km/s/Mpc}$$

$$h = 0.6736$$

$$\Omega_B h^2 = 0.02237$$

$$\Omega_c h^2 = 0.1200$$

$$n_s = 0.9649$$

$$\sigma_8 = 0.8111$$

$$A_s \approx 2.1 \times 10^{-9}$$

$$\Omega_M = 0.3153$$

$$\tau = 0.0544$$

$$T_{\text{CMB}} = 2.7255 \text{ K}$$

TT,TE,EE+lowE+lensing best-fit (Planck Collaboration 2020, arXiv:1807.06209).

Hubble Rate and Distances

$$E(z) = \sqrt{\Omega_{\text{M}}(1+z)^3 + \Omega_{\Lambda}} \quad (1)$$

$$H(z) = H_0 E(z) \quad (2)$$

$$\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad [\text{Mpc}/h] \quad (3)$$

$$d_L(z) = (1+z) \chi(z) \quad (4)$$

$$P_{\text{lin}}(k, z) \quad \text{via CAMB Boltzmann solver} \quad (5)$$

$$D(z) \propto E(z) \int_z^\infty \frac{(1+z') dz'}{E(z')^3}, \quad D(0) = 1 \quad (6)$$

$$\sigma^2(R, z) = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P_{\text{lin}}(k, z) W^2(kR) \quad (7)$$

$$W(x) = \frac{3(\sin x - x \cos x)}{x^3} \quad (\text{top-hat}) \quad (8)$$

Halo Model

Virial Overdensity and Halo Structure

Bryan & Norman (1998):

$$\Delta_{\text{vir}}(z) = 18\pi^2 + 82x - 39x^2, \quad x = \Omega_m(z) - 1 \quad (9)$$

Virial radius and circular velocity:

$$R_{\text{vir}} = \left(\frac{3M}{4\pi \Delta_{\text{vir}}(z) \rho_c(z)} \right)^{1/3} \quad [\text{Mpc}/h] \quad (10)$$

$$v_c = \sqrt{\frac{GM_{\text{phys}}}{R_{\text{phys}}}} \quad [\text{km/s}] \quad (11)$$

Physical units: $M_{\text{phys}} = M/h$, $R_{\text{phys}} = R_{\text{vir}}/h$.

Sheth–Mo–Tormen Mass Function and Bias

Peak height: $\nu = \delta_c^2 / \sigma^2(M, z)$, $\delta_c = 1.686$

Mass function (Sheth, Mo & Tormen 2001):

$$\nu f(\nu) = A [1 + (q\nu)^{-p}] \sqrt{\frac{q\nu}{2\pi}} \exp\left(-\frac{q\nu}{2}\right) \quad (12)$$

$q = 0.707$, $p = 0.3$, $A \approx 0.3222$

Halo bias (Sheth & Tormen 1999):

$$b(\nu) = 1 + \frac{q\nu - 1}{\delta_c} + \frac{2p}{\delta_c [1 + (q\nu)^p]} \quad (13)$$

Correa et al. (2015) concentration–mass:

$$\log_{10} c_{200} = \alpha + \beta \log_{10} \left(\frac{M}{M_{\odot}} \right) \left[1 + \gamma \left(\log_{10} \frac{M}{M_{\odot}} \right)^2 \right] \quad (14)$$

with z -dependent (α, β, γ) polynomials (Planck Appendix B1).

Converted $c_{200} \rightarrow c_{\text{vir}}$ via Newton iteration on the implicit NFW relation.

Analytic NFW Fourier transform:

$$\tilde{u}(k|M) = \frac{1}{f(c)} \left[\sin(kr_s)(\text{Si}_{1+c} - \text{Si}_1) + \cos(kr_s)(\text{Ci}_{1+c} - \text{Ci}_1) - \frac{\sin(ckr_s)}{(1+c)kr_s} \right] \quad (15)$$

where $f(c) = \ln(1+c) - c/(1+c)$ and $r_s = R_{\text{vir}}/c$.

HI Model

HI Mass–Halo Mass Relation

Padmanabhan, Refregier & Amara (2017), modified NFW fit:

$$M_{\text{HI}}(M, z) = \alpha f_{H,c} M \left(\frac{M}{10^{11} h^{-1} M_{\odot}} \right)^{\beta} \exp \left[- \left(\frac{v_{c,0}}{v_c(M, z)} \right)^3 \right] \quad (16)$$

α	β	$v_{c,0}$ [km/s]	$c_{\text{HI},0}$	γ_c
0.176	-0.69	40.7	139	0.13

$$f_{H,c} = (1 - Y_P) \Omega_B / \Omega_M, \quad Y_P = 0.24$$

HI concentration: $c_{\text{HI}} = c_{\text{HI},0} (M/10^{11} M_{\odot})^{-0.109} \cdot 4/(1+z)^{\gamma_c}$

Note: mass pivot is $10^{11} M_{\odot}$ (physical), not $M_{\odot}/h$.

Modified NFW HI Profile

$$\rho_{\text{HI}}(r) = \frac{\rho_0 r_s^3}{(r + 0.75 r_s)(r + r_s)^2} \quad (17)$$

Normalized: $\int_0^{R_{\text{vir}}} 4\pi r^2 \rho_{\text{HI}} dr = M_{\text{HI}}$

Analytic normalization via partial fractions ($A=9, B=-8, C=-4$):

$$\int_0^c \frac{x^2 dx}{(x+0.75)(x+1)^2} = 9 \ln \frac{c+0.75}{0.75} - 8 \ln(c+1) + 4 \left(\frac{1}{c+1} - 1 \right) \quad (18)$$

Fourier transform: Numerical quadrature

$$\tilde{u}_{\text{HI}}(k|M) = \frac{4\pi}{M_{\text{HI}}} \int_0^{R_{\text{vir}}} r^2 \rho_{\text{HI}}(r) \frac{\sin kr}{kr} dr \quad (19)$$

Brightness Temperature and HI Window

Mean comoving HI density and bias:

$$\bar{\rho}_{\text{HI}}(z) = \int \frac{dn}{dM} M_{\text{HI}}(M, z) dM \quad (20)$$

$$\Omega_{\text{HI}}(z) = \bar{\rho}_{\text{HI}}/\rho_{\text{c}} \quad (\text{D5: computed, not fixed}) \quad (21)$$

$$b_{\text{HI}}(z) = \frac{1}{\bar{\rho}_{\text{HI}}} \int \frac{dn}{dM} M_{\text{HI}} b(M, z) dM \quad (22)$$

Brightness temperature:

$$\bar{T}_b(z) = 188 h \Omega_{\text{HI}}(z) \frac{(1+z)^2}{E(z)} \text{ mK} \quad (23)$$

Window function (per- χ):

$$W_{\text{HI}}(\chi) = \bar{T}_b(z) \phi(z) \frac{H(z)}{c h}, \quad \phi(z) = \frac{1}{z_{\text{max}} - z_{\text{min}}} \quad (24)$$

Bias enters through P_{HI} , **not** through W_{HI} .

Cunnington Brightness Mode (D15)

Alternative for MeerKLASS data analyses (Cunnington et al. 2025):

$$\Omega_{\text{HI}}^{\text{C}}(z) = 6.7432 \times 10^{-4} + 3.9 \times 10^{-4} z - 6.5 \times 10^{-5} z^2 \quad (25)$$

$$b_{\text{HI}}^{\text{C}}(z) = 0.842 + 0.693 z - 0.0459 z^2 \quad (26)$$

$$\bar{T}_b^{\text{C}}(z) = 180 h \Omega_{\text{HI}}^{\text{C}}(z) \frac{(1+z)^2}{E(z)} \text{ mK} \quad (27)$$

Key differences from Padmanabhan mode:

- 180 mK prefactor (Battye et al. 2013) vs 188 mK
- Polynomial Ω_{HI} , b_{HI} vs halo-model integrals
- $P_{\text{HI}}^{2h} \rightarrow b_{\text{HI}}^2 P_{\text{lin}}$ (no halo integral)

Dispatched via per-telescope `default_hi_brightness` field.

Dark Matter Model

NFW ρ^2 Profile and Clumping Factor

Scale density and ρ^2 integral:

$$\rho_s = \frac{M}{4\pi r_s^3 f(c)} \quad (28)$$

$$\int_0^{R_{\text{vir}}} 4\pi r^2 \rho_{\text{NFW}}^2 dr = \frac{4\pi}{3} \rho_s^2 r_s^3 \left[1 - \frac{1}{(1+c)^3} \right] \quad (29)$$

Moliné et al. (2017) substructure boost:

$$\log_{10} B(M, z=0) = \sum_{i=0}^5 b_i \left[\log_{10} \frac{M}{M_{\odot}} \right]^i, \quad B(M, z) = \frac{B(M, 0)}{1+z} \quad (30)$$

Clumping factor (physical-frame definition):

$$\Delta^2(z) = \frac{1}{\bar{\rho}_m^2 (1+z)^3} \int \frac{dn}{dM} [1 + B(M, z)] \int \rho^2 d^3x dM \quad (31)$$

Why the $(1+z)^3$ factors?

1. **Clumping:** Halo integral gives comoving $\langle \rho^2 \rangle_{\text{com}}$.
Divide by $\bar{\rho}_{\text{com}}^2 \cdot (1+z)^3$ to get physical Δ_{phys}^2 .
2. **Window function:** Annihilation rate $\propto \rho_{\text{phys}}^2 = \rho_{\text{com}}^2 \cdot (1+z)^6$.
Per comoving volume: divide by $(1+z)^3$.
Net factor in W_{γ}^{DM} : $(1+z)^3$.
3. **End-to-end:** $\frac{\Delta^2}{(1+z)^3} \times (1+z)^3$ in W_{γ}^{DM} recovers the comoving integral.

The factors do **not** simply cancel — they serve different physical roles.

DM Annihilation Window Function

$$W_{\gamma}^{\text{DM}}(\chi) = \frac{\langle \sigma v \rangle}{8\pi} \left(\frac{\rho_{\text{DM}}}{m_{\chi}} \right)^2 (1+z)^3 \Delta^2(z) \frac{dN}{dE'} e^{-\tau(E,z)} \quad (32)$$

- $E' = (1+z) E_{\text{obs}}$ (rest-frame for PPC spectrum)
- $\tau(E_{\text{obs}}, z)$ (EBL at **observed** energy)
- No baked-in $1/H(z)$ — Limber measure supplies $d\chi/dz$
- $\rho_{\text{DM}} = \Omega_{\text{DM}} \cdot \rho_c$ (comoving, converted to GeV/cm^3)

Units: $[\text{photons (Mpc}/h)^{-3} \text{ s}^{-1} \text{ sr}^{-1} \text{ GeV}^{-1}]$

Astrophysical Sources

Four Source Populations

Source	GLF method	α (photon index)	Reference
BL Lac	LDDE inverse-sum	2.11	Ajello+2014
FSRQ	LDDE inverse-sum	2.44	Ajello+2012
mAGN	Radio→gamma chain	2.37	Di Mauro+2014
SFG	IR→gamma chain	2.70	Gruppioni+2013

All share the same window function structure $W_{\gamma}^{\text{astro}}$, differing only in their GLF $\Phi(L, z)$.

LDDE Luminosity Function (Blazars)

Double power-law:

$$\frac{d\Phi}{d \log_{10} L} = \frac{A}{(L/L_c)^{\gamma_1} + (L/L_c)^{\gamma_2}} \quad (33)$$

Smooth inverse-sum evolution (Ajello+2012 Eq. 15, D12):

$$e(z, L) = [r^{-p_1} + r^{-p_2}]^{-1}, \quad r = \frac{1+z}{1+z_c(L)} \quad (34)$$

with $z_c(L) = z_c^* (L/L_{\text{ref}})^\alpha$.

	A	L_c [erg/s]	γ_1	γ_2	z_c^*	p_1	p_2
FSRQ	3.06×10^{-9}	8.4×10^{47}	0.21	1.58	1.47	7.35	-6.51
BL Lac	9.20×10^{-11}	2.43×10^{48}	1.12	3.71	1.67	4.50	-12.88

4-step chain (Di Mauro+2014 / Pinetti 2020 Eq. C.19):

1. $L_\gamma \rightarrow \nu L_{\nu, \text{core}}^{5\text{GHz}}$: $\log L_\gamma = 2.0 + 1.008 \log(\nu L_\nu)$ [erg/s]
2. $\nu L_\nu \rightarrow L_{\text{core}}^{5\text{GHz}}$: divide by $\nu \cdot 10^7$ [W/Hz]
3. $L_{\text{core}} \rightarrow L_{\text{tot}}^{1.4\text{GHz}}$: Lara et al. (2004): $\log L_{\text{core}} = 4.2 + 0.77 \log L_{\text{tot}}$
4. $L_{\text{tot}}^{1.4} \rightarrow L_{151}$: $(1400/151)^{-0.80}$ [Inoue 2011]

Assembled GLF:

$$\phi_\gamma = \frac{k \eta(z)}{(1+z)^{2-\Gamma}} \frac{\rho_r(L_{151}, z)}{\ln 10 \cdot L_{151}} \left| \frac{dL_{151}}{dL_\gamma} \right| \quad (35)$$

$k = 3.05$, $\Gamma = 2.37$. $\eta(z)$ = Willott→Planck volume correction.

Gruppioni+2013 modified Schechter (3 components):

$$\phi_i = \phi_{0,i}(z) \left(\frac{L_{\text{IR}}}{L_{0,i}} \right)^{1-\gamma_i} \exp \left[-\frac{\log_{10}^2(1 + L_{\text{IR}}/L_{0,i})}{2\sigma_i^2} \right] \quad (36)$$

$$\phi_{\text{IR}} = \phi_{\text{spiral}} + \phi_{\text{starburst}} + \phi_{\text{SF-AGN}}$$

Ackermann+2012 L_γ - L_{IR} scaling:

$$\log_{10} L_\gamma = 1.09 \log_{10} \left(\frac{L_{\text{IR}}}{10^{10} L_\odot} \right) + 39.19 \quad (37)$$

SFG GLF:

$$\phi_\gamma^{\text{SFG}} = \phi_{\text{IR}} \cdot \frac{|d \log L_{\text{IR}} / d \log L_\gamma|}{L_\gamma \ln 10} \quad (38)$$

Astrophysical Window Function

$$W_{\gamma}^{\text{astro}}(\chi) = \frac{1}{4\pi h^3} \int_{L_{\min}}^{L_{\text{up}}} \Phi(L, z) \frac{L}{E_{\text{GeV} \rightarrow \text{erg}} I_{\alpha}} E_{\text{rest}}^{-\alpha} dL \times e^{-\tau(E, z)} \quad (39)$$

- $E_{\text{rest}} = (1 + z) E_{\text{obs}}$ (K-correction inside window)
- $1/h^3$ converts Mpc^{-3} to $(\text{Mpc}/h)^{-3}$
- $L_{\text{up}} = \min(L_{\text{max}}, L_{\text{sens}}(z))$ for unresolved sources
- D14: EBL attenuation $e^{-\tau}$ at observed energy

F_{sens} **dispatch (D17):**

- 'pinetti_constant': $F_{\text{sens}} = 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ (Pinetti+2020)
- '4fgl_dr4_psf': $F_{\text{sens}} = 7.3 \times 10^{-11} \text{ cm}^{-2} \text{ s}^{-1}$ (Ballet+2023) + PSF scaling

EBL and PPC4DMID

Extragalactic Background Light Opacity

Dominguez et al. (2011) tabulated opacity via ebltable:

$$\tau(E_{\text{obs}}, z) \quad (\text{pair-production optical depth}) \quad (40)$$

$$A(E_{\text{obs}}, z) = e^{-\tau(E_{\text{obs}}, z)} \quad (\text{attenuation factor}) \quad (41)$$

Key convention: E is the **observed** photon energy (at Earth), not rest-frame.

Available models: dominguez (default), franceschini, finke, saldana-lopez21.

Where EBL enters:

- W_{γ}^{DM} : attenuates DM annihilation photons
- $W_{\gamma}^{\text{astro}}$: attenuates astrophysical source photons (D14)
- L_{sens} : raises detection threshold at high z via $J_{\alpha}^{\text{EBL}}(z)$

Cirelli et al. (2011): Tabulated $dN/d \log_{10} x$ per annihilation event.

$$x = E/m_\chi \tag{42}$$

$$\frac{dN}{dx} = \frac{dN/d \log_{10} x}{x \ln 10} \tag{43}$$

$$\frac{dN}{dE} = \frac{1}{m_\chi} \frac{dN}{dx} \tag{44}$$

Channels used: $b\bar{b}$ (primary), $\tau^+\tau^-$, W^+W^- , ZZ , $c\bar{c}$, $t\bar{t}$, gg , hh , etc.

Mass range: 5 GeV – 100 TeV.

D7: Pipeline uses public PPC4DMID tables (thesis used private Pythia code).

Noise and Beams

Radio System Temperature

Pinetti generic (Camera et al. 2013; legacy MeerKAT/SKA):

$$T_{\text{sys}}^{\text{P}}(\nu) = 30 \text{ K} + 60 \text{ K} \left(\frac{300 \text{ MHz}}{\nu} \right)^{2.55} \quad (45)$$

MeerKAT canonical (Cunnington et al. 2025 Eq. A19; D16):

$$T_{\text{sys}}^{\text{M}}(\nu) = T_{\text{rx}}(\nu) + T_{\text{spl}} + T_{\text{CMB}} + T_{\text{gal}}(\nu) \quad (46)$$

$$T_{\text{rx}} = 7.5 + 10 (\nu/\text{GHz} - 0.75)^2 \text{ K}$$

$$T_{\text{spl}} = 3 \text{ K}, \quad T_{\text{CMB}} = 2.725 \text{ K}$$

$$T_{\text{gal}} = 15 \text{ K} (408 \text{ MHz}/\nu)^{2.75}$$

Gives $\sim 14 \text{ K}$ (L-band), $\sim 16 \text{ K}$ (UHF) — half the Pinetti formula.

Dispatched via `T_sys_model`: 'pinetti' or 'meerkat'.

Single-dish:

$$N_{\text{dish}} = \frac{T_{\text{sys}}^2 \Omega_S}{N_d t \Delta\nu N_b N_{\text{pol}} \eta^2} \quad [\text{mK}^2 \text{ sr}] \quad (47)$$

Interferometer:

$$N_{\text{interf}}(\ell) = \frac{T_{\text{sys}}^2 \Omega_S \text{FoV}}{n(u) t \Delta\nu N_b N_{\text{pol}} \eta^2} \quad (48)$$

Valid for $\ell \geq \ell_{\text{cut}} = \pi D_{\text{short}} / (1.22 \lambda)$

Combined: $N_{\ell}^{\text{radio}} = \min(N_{\text{dish}}, N_{\text{interf}})$ for $\ell \geq \ell_{\text{cut}}$

Beam: $B_{\ell}^{\text{HI}} = \exp(-\ell^2 \sigma_{\text{beam}}^2 / 2)$, $\sigma_{\text{beam}} = 1.22 \lambda / (D \sqrt{8 \ln 2})$

Gaussian approximation (forecast mode, Pinetti+2020):

$$\sigma_0(E) = 1.20 (E/0.5 \text{ GeV})^{-0.95} + 0.05 \quad (49)$$

$$\sigma_b(\ell) = \sigma_0 / (1 + 0.25 \sigma_0 \ell) \quad (50)$$

$$B_\ell^\gamma = \exp(-\sigma_b^2 \ell^2 / 2) \quad (51)$$

Exact King PSF (data mode, Ammazzalorso+2018):

$$\text{PSF}(\theta) = \frac{1 - 1/\gamma}{2\pi\sigma_K^2} \left[1 + \frac{\theta^2}{2\gamma\sigma_K^2} \right]^{-\gamma}, \quad \gamma = 2 \quad (52)$$

$$W_\ell(E) = 2\pi \int_0^\pi \text{PSF}(\theta) P_\ell(\cos \theta) \sin \theta d\theta \quad (53)$$

Energy-averaged: $\langle W_\ell^k \rangle = \int W_\ell(E) E^{-\alpha} dE / \int E^{-\alpha} dE, \quad \alpha = 2.3$

Limber Integration

Cross-Power Spectra (Halo Model)

HI $\times$ DM 2-halo (Pinetti Eq. 5.2):

$$P_{\text{HI} \times \text{DM}}^{2h}(k, z) = \underbrace{\left[\int \frac{dn}{dM} b \frac{\tilde{v}}{\Delta^2} dM \right]}_{\text{DM kernel}} \underbrace{\left[\int \frac{dn}{dM} b \frac{\tilde{u}_{\text{HI}} M_{\text{HI}}}{\bar{\rho}_{\text{HI}}} dM \right]}_{\text{HI kernel}} P_{\text{lin}}(k, z) \quad (54)$$

HI $\times$ Astro 2-halo (Pinetti Eq. 5.4):

$$P_{\text{HI} \times \text{astro}}^{2h}(k, z) = \left[\int \frac{dn}{dM} b \frac{\tilde{u}_{\text{HI}} M_{\text{HI}}}{\bar{\rho}_{\text{HI}}} dM \right] b_{\text{astro}} P_{\text{lin}}(k, z) \quad (55)$$

Cunnington mode: HI kernel $\rightarrow b_{\text{HI}}^{\text{C}}(z)$ (polynomial), no mass integral.

$$C_{\ell}^{ij} = \int \frac{d\chi}{\chi^2} W_i(\chi) W_j(\chi) P_{ij} \left(k = \frac{\ell + \frac{1}{2}}{\chi}, z \right) \quad (56)$$

D8: $k = (\ell + \frac{1}{2})/\chi$ (LoVerde & Afshordi 2008) vs thesis $k = \ell/\chi$.

Implementation:

- Riemann sum over redshift grid ($n_z = 200$ points in $[z_{\min}, z_{\max}]$)
- Weight: $(d\chi/dz)/\chi^2 \cdot \Delta z$ where $d\chi/dz = c h/H(z)$ [Mpc/h]
- Product: $W_{\text{HI}} \times W_{\gamma} \times P_{\text{cross}}(k, z) \times \text{weight}$

Multi-energy optimisation: The kernel $(\chi, H, W_{\text{HI}}, P_{\text{cross}})$ is E -independent. Only $W_{\gamma}(E, z)$ varies with energy. Compute kernel once, loop over E in inner loop.

Dimensional Analysis of C_ℓ

Factor	Units
$d\chi/dz$	[Mpc/h]
$1/\chi^2$	[(Mpc/h) ⁻²]
W_{HI}	[mK · (Mpc/h) ⁻¹]
W_γ	[ph (Mpc/h) ⁻³ s ⁻¹ sr ⁻¹ GeV ⁻¹]
P_{cross}	[(Mpc/h) ³]
dz	[1]
$C_\ell^{\text{HI} \times \gamma}$	[mK · ph · (Mpc/h) ⁻² · s ⁻¹ · sr ⁻¹ · GeV ⁻¹]

The (Mpc/h)⁻² is the angular projection factor.

Converting to cm gives the standard cross-power units.

Statistics

Noise-dominated approximation (Pinetti Eq. 5.5):

$$(\Delta C_\ell)^2 = \frac{1}{(2\ell + 1) f_{\text{sky}}} \frac{N^\gamma}{(B_\ell^\gamma)^2} \left[C_\ell^{\text{HI,HI}} + \frac{N^{\text{HI}}}{(B_\ell^{\text{HI}})^2} \right] \quad (57)$$

Unit conversion: $N^\gamma [\text{cm}^{-4} \text{s}^{-2} \text{sr}^{-1}] \rightarrow [(\text{Mpc}/h)^{-4} \text{s}^{-2} \text{sr}^{-1}]$ via $\times (L_{\text{Mpc}/h}^{\text{cm}})^4$

Effective sky fraction: $f_{\text{sky}} = \min(f_{\text{sky}}^{\text{radio}}, f_{\text{sky}}^\gamma)$

$$\text{SNR}^2 = \sum_{\ell, E\text{-bins}} \left[\frac{C_{\ell}^{\text{HI} \times \gamma_{\star}}}{\Delta C_{\ell}} \right]^2 \quad (58)$$

- Numerator: astrophysical-only signal (sum of 4 source classes)
- Denominator: noise-dominated variance
- Sum over 12 Fermi energy bins (forecast) or 11 Ammazzalorso+2018 bins (data)
- ℓ -range: $[10, 1000]$ (forecast) or per-bin $[\ell_{\min}, \ell_{\max}(E)]$ (data)

Per-telescope dispatch: `compute_SNR` reads `default_hi_brightness` and `default_unresolved_mode` from the telescope config entry.

DM Exclusion Curves

$\Delta\chi^2$ test statistic (Pinetti Eq. 5.7):

$$\Delta\chi^2 = \sum_{\ell, E} \left[\frac{C_{\ell}^{\text{castro+DM}}}{\Delta C_{\ell}} \right]^2 - \left[\frac{C_{\ell}^{\text{castro}}}{\Delta C_{\ell}} \right]^2 \quad (59)$$

Exclusion scaling: Since $C_{\ell}^{\text{DM}} \propto \langle\sigma v\rangle$:

$$\langle\sigma v\rangle^{\text{excl}} = \langle\sigma v\rangle^{\text{test}} \sqrt{\frac{\Delta\chi_{\text{threshold}}^2}{\Delta\chi_{\text{test}}^2}} \quad (60)$$

- 95% CL: $\Delta\chi_{\text{thr}}^2 = 4.0$
- 5σ : $\Delta\chi_{\text{thr}}^2 = 25.0$

Scan over $m_{\chi} \in [10 \text{ GeV}, 10 \text{ TeV}]$ to produce the exclusion curve.

Deviations from Literature

Active Deviations Registry

#	Deviation	Nature
D2	Correa $c(M, z)$ coefficients	Planck App. B1 fit (not thesis cosmology)
D4	SF-AGN k_{R2} sign	Thesis typo corrected (-3.17 , not $+3.17$)
D5	Ω_{HI} computed	Halo integral vs fixed 2.45×10^{-4}
D6	SMT $q = 0.707$	Original SMT 1999; thesis uses $q = 0.75$
D7	PPPC4DMID public tables	vs thesis private Pythia code
D8	Limber $k = (\ell + \frac{1}{2})/\chi$	LoVerde & Afshordi 2008 correction
D12	LDDE exponent form	Ajello+2012 positive-exponent convention
D14	EBL on astro windows	Now applied; thesis omitted for astro
D15	Cunnington brightness mode	180 mK + polynomial Ω_{HI} , b_{HI}
D16	MeerKAT T_{sys}	15 K Galactic synchrotron (Cunnington+2025)
D17	4FGL-DR4 F_{sens} dispatch	7.3×10^{-11} + PSF scaling (Ballet+2023)

Resolved Deviations

#	Was	Resolution
D3	No $c_{200} \rightarrow c_{\text{vir}}$ conversion	<code>c200_to_cvir()</code> now implemented
D9	$\bar{T}_b$ 188h vs 44 μK form	Mathematically equivalent (convention)
D11	$\Delta_{\text{vir}} = 200$ fixed	Bryan & Norman z -dependent $\Delta_{\text{vir}}(z)$
D13	Gruppioni L_0 break uniform	Now spiral-only per Table 8

Cosmology Mismatch Registry

Fit	Paper cosmology	Correction	Residual
Correa+2015 $c(M)$	Planck 2013	D2: re-fit	Negligible
Ajello+2012 FSRQ	WMAP-era ($h=0.71$)	None	$\sim 1\text{--}2\%$ in d_L
Ajello+2014 BL Lac	WMAP-era ($h=0.71$)	None	$\sim 1\text{--}2\%$ in d_L
Willott+2001	$H_0=50$, empty	$\eta(z)$ volume	Shape not re-derived
Gruppioni+2013	$H_0=70$, $\Omega_m=0.3$	None	Small
Moliné+2017	Planck 2015	None	Weak (N-body)
Padmanabhan+2017	WMAP-3 custom	None	Weak
Cunnington+2025	Planck 2018	None needed	—
Ballet+2023	Cosmology-independent	N/A	—

Willott+2001 $\eta(z)$ corrects density only — luminosity rescaling would require re-fitting to original flux data (documented limitation from Di Mauro+2014).